Retail sales Analysis

Project Overview

- *This project focuses on analyzing sales data using Microsoft Excel to evaluate business performance, identify trends, and generate actionable insights.*
- *The analysis includes yearly sales comparison, sales distribution, and performance evaluation using Pivot Tables and visualizations.*

Objective

- *To analyze and summarize sales data effectively*
- *To perform **annual sales comparison** and identify growth trends*
- *To extract meaningful insights for better decision-making*
- *To visualize sales performance using Pivot Tables and charts*

Data Sources

- ***Source Description:** The dataset was extracted from datasetsearch.google.com for the purpose of academic analysis.*
- ***Timeline:** 2023-2024*
- ***Domain:** Retail sales*

Problem Statement

• To analyze overall sales performance across different time periods.

• To compare annual sales and identify growth or decline trends.

• To evaluate sales distribution across categories/segments.

• To identify patterns and variations in sales data.

• To transform raw data into meaningful insights for decision-making.

Attribute (Column /Features) Details:

Attribute Name	*Data Type*	*Description*
Order ID	*General*	*Unique number for each order*
Customer ID	*General*	*Unique idendifier for each customer*
Product Category	*Text*	*Classification of products into groups*
Product name	*Text*	*Name of the specific product purchased by the customer.*
Quantity	*Number*	*Number of units of a product purchased in a single order.*
Unit price	*Currency*	*Price of a single unit of the product before applying any discount.*
Discount	*Number*	*Reduction applied to the product price, usually expressed as a percentage or amount.*
Total sales	*Currency*	*Final sales amount after considering quantity and discount.*
Order date	*Date*	*Date on which the order was placed.*
Region	*Text*	*Geographical area where the order was placed or delivered*
Payment method	*Text*	*Mode of payment used by the customer*

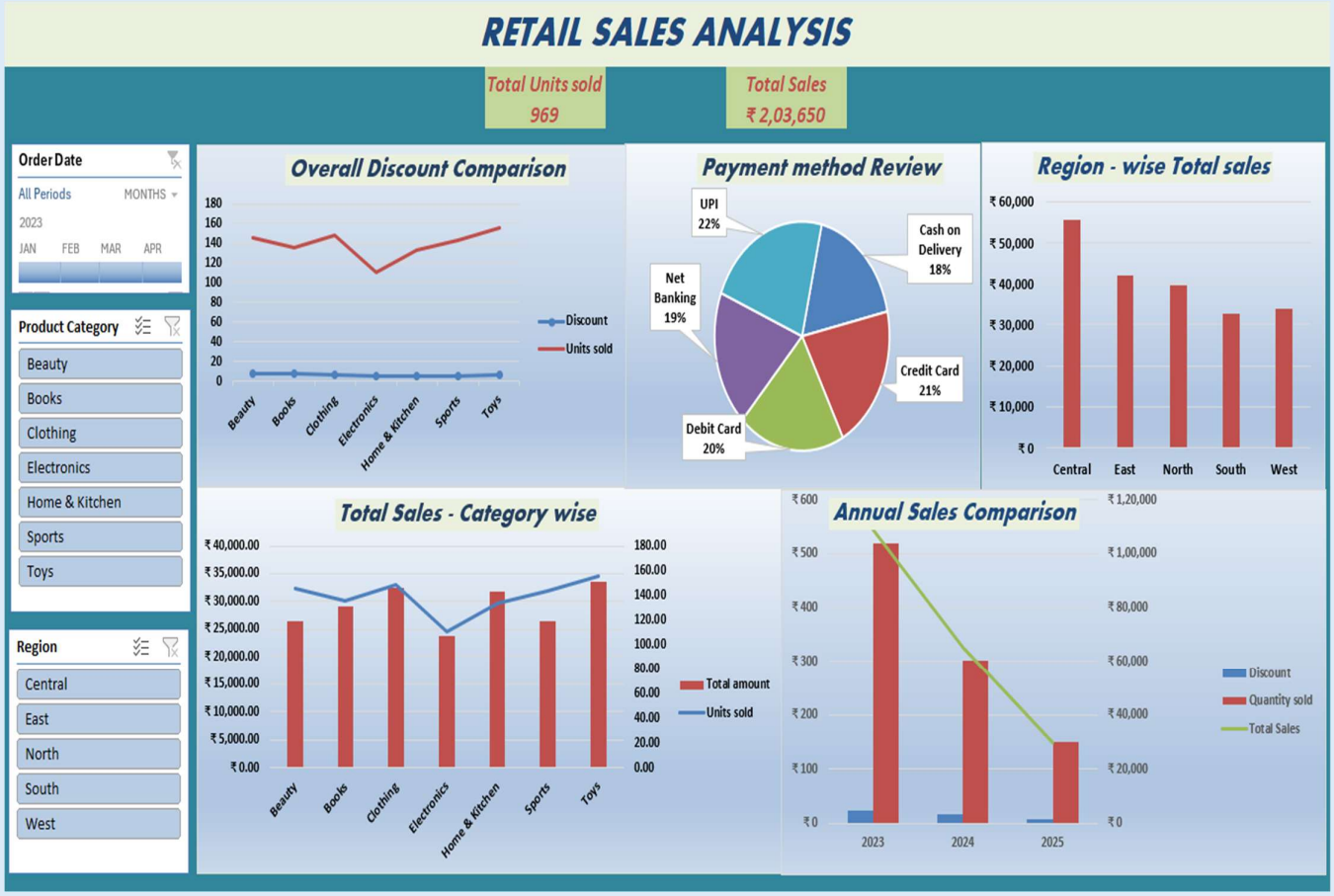
Tools & Technologies

- *Microsoft Excel*
- *Pivot Tables*
- *Pivot Charts*
- *Data Cleaning Techniques*
- *Excel Functions*
- *Visualization*

Data Pre-Processing

- *Cleaned and removed inconsistent or missing values*
- *Converted date column into proper date format*
- *Extracted **Year** from date column for time-based analysis*
- *Structured dataset for Pivot Table creation*
- *Verified data accuracy and consistency*

Visualization



Insights

- *Total of 969 units sold generating ₹2,03,650 revenue*
- *Indicates steady sales performance with scope for growth*
- *Revenue is moderately strong relative to volume*
- *Categories like Clothing and Toys show higher total sales*
- *Electronics has comparatively lower performance*
- *Units sold vary across categories → demand is not uniform*
- *Discounts are consistent across categories*
- *No heavy discount spikes → controlled pricing strategy*
- *UPI is the most preferred payment method (~22%)*
- *Followed by Credit Card, Debit Card, and Net Banking*
- *Cash on Delivery is the least used*
- *Central region generates the highest sales*
- *South and West regions are lower in comparison*
- *Sales show a declining trend from 2023 to 2025*
- *Quantity sold also decreases over time*

Conclusion

- *Increase focus on **low-performing regions***
- *Boost sales in weaker categories like **Electronics***
- *Introduce **targeted discounts or offers***
- *Strengthen **customer engagement and retention strategies***
- *Analyze reasons for **yearly decline in sales***
- *The analysis helps in understanding sales performance, identifying growth trends, and supporting data-driven decision-making.*

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